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localhost:8888/notebooks/GEN%20AI%20(JY)%2Fass_18.ipynb

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JupyterLab Python 3 (ipykernel)

```
[1]: import numpy as np
import pandas as pd
ads = np.array([[2], [4], [6], [8], [10], [12]])
sales = np.array([20, 35, 50, 65, 80, 95])
print("Sales Data")
data = pd.DataFrame({
    "Advertisements": ads.flatten(),
    "Sales": sales
})
print(data)
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(ads, sales)
new_ads = np.array([[7]])
prediction = model.predict(new_ads)
print("Predicted Sales:", prediction[0])
import matplotlib.pyplot as plt
plt.scatter(ads, sales, label="Actual Data")
plt.plot(ads, model.predict(ads), label="Regression Line")
plt.xlabel("Advertisements")
plt.ylabel("Sales")
plt.title("Sales Prediction")
plt.legend()
plt.show()
```

```
Sales Data
Advertisements Sales
0      2      20
1      4      35
2      6      50
3      8      65
4     10      80
5     12      95
Predicted Sales: 57.5
```

